

DTP

Debug and Test Ports

Beta release | **Open-source RTL** | **Apache 2.0**

HIGHLIGHTS

- JTAG TAP and boundary-scan access implementing IEEE 1149.1-2013; the boundary-scan network and logic are supplied by adopter DFT tooling and pad IP
- iJTAG instrument networks implementing IEEE 1687-2014
- Configurable secondary TAP topology, compliant with IEEE 1838, for local and downstream chiplet access
- JTAG-to-AXI paths for SMC fabric, SMC OTP, and SEP OTP debug access
- Cross-trigger routing between chiplet-local and external trigger endpoints
- Lifecycle-controlled disable inputs for all internal and external debug access ports
- Integrated debug clock-stop, boot-stall, and multi-target reset controls

SYSTEM ROLE

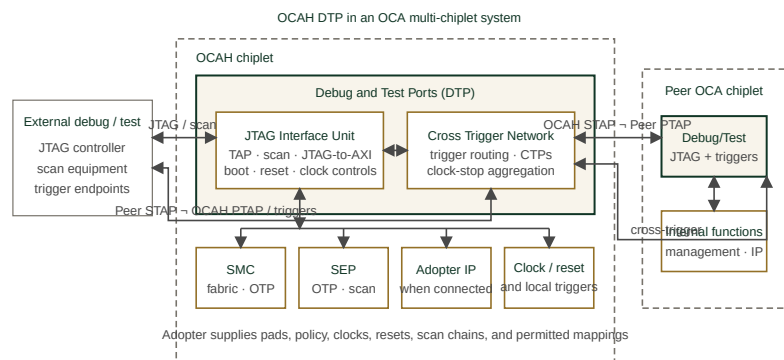
DTP sits between external debug and test equipment, OCAH management and security functions, adopter IP, and neighboring chiplets. It is a component of the OCAH SMU, a first-class subsystem delivered as part of OCAH whose reference components include the subsystem wrappers and IP-integration modules. DTP can also form a subsystem-level boundary when the adopter supplies its lifecycle policy, fabric, reset, clock, scan, and pad connections.

OVERVIEW

The Debug and Test Ports (DTP) subsystem provides the central debug and test infrastructure for an OCAH chiplet. Its JTAG Interface Unit implements interfaces based on IEEE 1149.1-2013, IEEE 1687-2014, and IEEE 1838. Its Cross Trigger Network coordinates local and inter-chiplet triggers. Together, the two blocks provide controlled debug access, boundary scan, cross triggering, boot and reset controls, and clock control for multi-chiplet debugging.

A primary JTAG TAP provides boundary-scan, iJTAG, secondary-TAP, debug-control, capability, and JTAG-to-AXI data registers. Optional bridges issue debug transactions onto the SMC system fabric and the SMC and SEP OTP AXI4-Lite interfaces. Whether any adopter IP is reachable depends on the adopter's downstream address map. Lifecycle-derived, active-high disable fields are synchronized into the scan domain and gate the corresponding paths. When DTP is used standalone, the adopter owns the policy source and permitted address ranges; within the OCAH SMU, the SEP defines the lifecycle control policy.

SYSTEM CONTEXT



AT A GLANCE

Feature	Description
External / internal triggers	16 CTPs / 10 internal interfaces
Clock-stop request inputs	9
Cross-trigger CSR port	32-bit address, 32-bit data AXI4-Lite
SMC debug manager	56-bit address, 64-bit data AXI4
OTP debug managers	32-bit address, 32-bit data AXI4-Lite
Primary instruction register	6 bits

Values are the reference defaults; the CTP (1-32), internal trigger interface (0-32), and clock-stop request input (1-32) counts are parameterizable. The default SMU uses two of the 10 internal trigger interfaces and one of the nine clock-stop request inputs for SMC connections. Its external boundary therefore exposes eight internal trigger interfaces and eight clock-stop request inputs; all 16 CTPs remain externally visible.

CAPABILITIES

JTAG and scan

- Primary TAP with IDCODE and BYPASS; SAMPLE/PRELOAD and EXTEST are compile-time optional with boundary scan enabled
- Compile-time optional INTEST, CLAMP, HIGHZ, RUNBIST, EXTEST_PULSE, EXTEST_TRAIN, TMP, and CLAMP hold/release functions
- Boundary-scan host interface plus separate DFD, secure DFT, non-secure DFT, extended STAP, I/O STAP, SMC STAP, SEP STAP, and extra-STAP connections
- Capability registers report enabled functions and JTAG-to-AXI limits

Debug and control

- Single and series JTAG-to-AXI transactions, including incrementing and fixed-address series operations and target error reporting
- Independent lifecycle disable fields for three bridge paths and eight scan-side paths
- Independently enabled reset-control outputs for SMC, SEP, and adopter logic
- Debugger-controlled boot-stall and functional clock-stop requests

Cross trigger

- Bidirectional external CTP signaling and chiplet-local trigger interfaces
- Software-controlled source/destination routing with point-to-point and wire-OR behavior
- Pulse synchronization or four-phase request/acknowledge mode per internal trigger interface

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	System clock/reset, JTAG TCK/TRST_N, JTAG power-on reset, scan reset, test enable
Lifecycle policy	Active-high, per-path debug and test disable controls
Scan	PTAP pins; BSR/iJTAG scan controls; I/O, SMC, SEP, and extra STAPs
Fabric	One AXI4 manager, two AXI4-Lite managers, one AXI4-Lite subordinate
Cross trigger	External pad controls, internal source/sink interfaces, clock-stop requests and output
Build-time	Instruction/path enables, extra STAP count, IDCODE/version fields, reset-control output types, trigger protocol modes, AXI type parameters

Build-time parameters select the boundary and internal feature set. Optional JTAG instructions are individually enabled (JTAG_BSR_ENABLE, JTAG_INTEST_ENABLE, JTAG_CLAMP_ENABLE, JTAG_HIGHZ_ENABLE, JTAG_RUNBIST_ENABLE, JTAG_TMP_ENABLE, and the EXTEST_PULSE/EXTEST_TRAIN enables); bridge and STAP paths are enabled per target (JTAG_SMC_DBG_ENABLE, JTAG_SEP_DBG_ENABLE, JTAG_STAP_IO_ENABLE, JTAG_NUM_EXTRA_STAPS); reset-control slices are enabled per domain (JTAG_IC_RESET_SMC_ENABLE, JTAG_IC_RESET_SEP_ENABLE, JTAG_IC_RESET_EXT_ENABLE); and identity, trigger protocol, and AXI types are set through the IDCODE, XTRIG_INT_CT_MODE, and AXI type parameters.

INTEGRATION DEPENDENCIES

- Generate lifecycle debug-disable policy outside DTP and connect the current active-high debug-disable contract.
- Replace the zero-valued IDCODE manufacturer, part, revision, and OCAH version defaults with release-specific values.
- Provide safe clock, reset, CDC, pad, and clock-gating integration. Establish project-specific TCK constraints; this beta sheet does not specify a characterized operating frequency.
- Configure the SMC, SEP, and adopter reset-control output types and enables, and connect or correctly bypass every enabled scan and secondary-TAP chain.
- Connect enabled AXI manager ports only to address maps permitted by the product security policy. Disabled and unused interfaces require documented tie-offs.
- Integrate the cross-trigger CSR subordinate into the management address map and connect the required local, external, and clock-stop endpoints.

VERIFICATION AND MATURITY

The OCAH [verification dashboard](#) publishes the current verification and maturity status. The repository’s open-source PyUVM environment exercises TAP operation, standard and optional instructions, JTAG-to-AXI data and error paths, backpressure/reset cases, iJTAG and STAP routing, cross-trigger modes, clock controls, and lifecycle-disable matrices. Scoreboards, reference models, coverage intent, and deliberately corrupted-checker tests guard against vacuous passes.

No third-party standards certification, silicon characterization, or portable frequency, area, and power figures are claimed. Those results depend on the selected configuration, libraries, constraints, physical implementation, and target process.

DELIVERABLES

- SystemVerilog RTL and dependency filelists
- SystemRDL-derived cross-trigger register collateral
- Integration, architecture, interface, and verification-plan documentation
- Open-source cocotb/PyUVM tests, reusable JTAG/AXI agents, scoreboards, reference models, and functional coverage
- Lint collateral and lint, CDC, and RDC waivers
- Synthesis and timing-constraint starting points
- Apache License 2.0 repository distribution

RESOURCES

Documentation: tenstorrent.github.io/tt-oca-harness
DTP source documentation: [hw/sys/dtp/doc/](#)
RTL: [hw/sys/dtp/rtl/](#)
Verification: [hw/sys/dtp/dv/](#)
Synthesis collateral: [hw/sys/dtp/synth/](#)
Integrator Guide: [doc/integrator/](#)
Project: github.com/tenstorrent/tt-oca-harness

TERMS

CTP: cross trigger port. DFD: design-for-debug. DFT: design for test. OTP: one-time programmable. PTAP: primary TAP. STAP: secondary TAP. CSR: control and status register.

DOCUMENT CONTROL

Version	Date	Author	Change